**FORM 2**

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COMPLETE SPECIFICATION

(See sections 10; rule 13)

**TITLE OF THE INVENTION**

SYSTEM AND METHOD FOR DYNAMIC CYBER DNA FINGERPRINTING AND EVOLUTIONARY ANALYSIS OF CYBERCRIME COMPLAINTS

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PREAMBLE TO THE DESCRIPTION

The following specification particularly describes the invention and the manner in which it is to be performed.

# SYSTEM AND METHOD FOR DYNAMIC CYBER DNA FINGERPRINTING AND EVOLUTIONARY ANALYSIS OF CYBERCRIME COMPLAINTS

## BACKGROUND

### *Technical Field*

[0001] The embodiments herein generally relate to the field of cybersecurity, artificial intelligence, cybercrime intelligence, and digital forensics, and more particularly, to a system and method for analysing cybercrime complaints using Dynamic Cyber DNA fingerprinting and evolutionary analysis of cybercrime patterns.

### *Description of the Related Art*

[0002] Cybercrime complaints are increasing with the widespread use of digital payments, social media, online banking, messaging platforms, and other internet-based services. Conventional cybercrime complaint systems generally focus on collecting complaints, categorizing reported incidents, storing digital evidence, and assisting authorities in reviewing individual cases. Although artificial intelligence and machine learning techniques have been introduced for fraud detection, complaint classification, entity extraction, and relationship analysis, existing approaches commonly analyse complaints as independent incidents or identify similarities based on predefined features, keywords, crime categories, or known entities.

[0003] Furthermore, cybercrime complaints may include heterogeneous digital evidence, such as complaint narratives, screenshots, messages, emails, chat records, transaction records, URLs, documents, QR codes, bank statements, and audio recordings. Information relevant to an investigation may be distributed across multiple evidence sources, making it difficult for conventional systems to generate a unified and reliable representation of the underlying cybercrime behaviour.

[0004] In addition, conventional approaches that rely primarily on direct entity matching or static feature comparison may fail to identify relationships between cybercrime incidents when attackers modify observable characteristics, such as communication channels, attack techniques, impersonation methods, or payment mechanisms. As a result, related cybercrime incidents may appear unrelated despite preserving an underlying behavioural strategy. Existing approaches may also lack mechanisms for systematically identifying behavioural characteristics that remain stable while operational characteristics change over time.

[0005] Further, conventional cybercrime analysis systems generally do not provide a dynamic complaint-level fingerprint capable of representing behavioural intent, attack sequence, communication characteristics, financial characteristics, temporal characteristics, infrastructure indicators, and evidence relationships in an interconnected manner. Such systems may also lack the ability to incrementally update the analytical representation when new evidence or validated investigative information becomes available, while preserving historical information and evidence provenance.

[0006] Accordingly, there exists a need for an improved system and method for analysing cybercrime complaints by generating dynamic behavioural fingerprints from multimodal evidence, identifying relationships between seemingly unrelated complaints, detecting changes and preserved behavioural characteristics, tracking evolutionary relationships among cybercrime patterns, and providing evidence-linked and explainable investigative intelligence. The embodiments described herein address these needs through Dynamic Cyber DNA fingerprinting, cross-complaint correlation, mutation analysis, evolutionary lineage analysis, campaign-level intelligence, and adaptive updating based on investigative feedback.

## SUMMARY

[0007] Embodiments herein provide a system for dynamic Cyber DNA fingerprinting and evolutionary analysis of cybercrime complaints. The system includes a memory configured to store cybercrime complaints, associated digital evidence, previously generated Cyber DNA fingerprints, and historical cybercrime relationship information, and one or more processors communicatively coupled to the memory. The one or more processors are configured to receive a cybercrime complaint and associated digital evidence comprising one or more of text, images, screenshots, messages, emails, chat records, transaction records, bank statements, URLs, QR codes, documents, and audio recordings; process the cybercrime complaint and the associated digital evidence to extract and normalize entities and cybercrime events and determine evidence support; determine behavioural intents and an attack sequence from the extracted cybercrime events; generate a Dynamic Cyber DNA fingerprint comprising interconnected genes representing intent characteristics, behavioural characteristics, communication characteristics, financial characteristics, temporal characteristics, infrastructure characteristics, and evidence characteristics; compare the Dynamic Cyber DNA fingerprint with previously generated Cyber DNA fingerprints based on direct entity relationships and behavioural similarity to identify

candidate relationships between cybercrime complaints; determine changes between corresponding characteristics and behavioural characteristics preserved despite the changes; generate a candidate Cyber DNA evolutionary lineage based on the determined changes, preserved behavioural characteristics, temporal information, and supporting evidence; and generate an evidence-linked investigation output identifying the candidate relationship, the determined changes, the preserved behavioural characteristics, and the candidate Cyber DNA evolutionary lineage.

[0008] In some embodiments, the one or more processors are configured to process each item of the associated digital evidence according to an evidence type of the item and generate a structured evidence intelligence record comprising normalized entities, cybercrime events, source provenance, extraction confidence, evidence agreement information, and conflicting evidence information.

[0009] In some embodiments, the one or more processors are configured to determine the behavioural intents by identifying respective micro-intents associated with the cybercrime events and arranging the respective micro-intents according to the attack sequence, each micro-intent being associated with corresponding supporting evidence and a confidence value based on semantic correspondence, evidence support, and contextual consistency.

[00010] In some embodiments, each gene of the Dynamic Cyber DNA fingerprint comprises a gene type, an encoded characteristic, an associated confidence value, a temporal position, and one or more relationships with other genes, thereby preserving a sequence and interrelationship of characteristics associated with the cybercrime complaint.

[00011] In some embodiments, the one or more processors are configured to update the Dynamic Cyber DNA fingerprint upon receipt of additional evidence or validated investigative information by identifying one or more genes affected by the additional evidence or validated investigative information, updating the identified genes while retaining unaffected genes, and maintaining a previous version and an updated version of the Dynamic Cyber DNA fingerprint with corresponding evidence provenance and update information.

[00012] In some embodiments, the candidate relationship is identified by determining an entity relationship score based on shared entities and an adaptive reliability weight associated with each entity type, determining a Cyber DNA similarity score based on similarities between

corresponding genes of the Dynamic Cyber DNA fingerprints, and combining the entity relationship score and the Cyber DNA similarity score to generate a hybrid relationship score.

[00013] In some embodiments, determining the changes and the preserved behavioural characteristics comprises identifying one or more characteristics that are preserved, modified, substituted, inserted, removed, or transformed between the Dynamic Cyber DNA fingerprints, and determining a behavioural core preservation measure and a context mutation measure to distinguish a candidate evolved variant from a merely similar cybercrime complaint.

[00014] In some embodiments, the one or more processors are further configured to store sufficiently supported or investigator-validated transformations between Cyber DNA characteristics in a Crime Evolution Memory, group related Dynamic Cyber DNA fingerprints into a candidate campaign profile, and estimate one or more candidate future variants based on previously stored transformations and contextual transition information, each candidate future variant being associated with supporting historical transition information and a probability or confidence value.

[00015] Embodiments herein further provide a computer-implemented method for dynamic Cyber DNA fingerprinting and evolutionary analysis of cybercrime complaints. The method includes receiving a cybercrime complaint and associated digital evidence; processing the cybercrime complaint and the associated digital evidence to extract and normalize entities and cybercrime events and determine evidence support; determining behavioural intents and an attack sequence from the cybercrime events; generating a Dynamic Cyber DNA fingerprint comprising interconnected genes representing intent characteristics, behavioural characteristics, communication characteristics, financial characteristics, temporal characteristics, infrastructure characteristics, and evidence characteristics; comparing the Dynamic Cyber DNA fingerprint with previously generated Cyber DNA fingerprints based on direct entity relationships and behavioural similarity to identify candidate relationships between cybercrime complaints; determining changes between corresponding characteristics of Dynamic Cyber DNA fingerprints associated with a candidate relationship and behavioural characteristics preserved despite the changes; generating a candidate Cyber DNA evolutionary lineage based on the changes, the preserved behavioural characteristics, temporal information, and supporting evidence; and generating an evidence-linked investigation output identifying the candidate relationship, the

changes, the preserved behavioural characteristics, and the candidate Cyber DNA evolutionary lineage.

[00016] In some embodiments, the method further comprises receiving investigator feedback indicating that a candidate relationship, mutation relationship, or campaign association is confirmed, rejected, or requires further investigation, and responsively updating one or more of entity reliability weights, Cyber DNA similarity parameters, mutation relationships, evolutionary lineage assessment parameters, and stored historical transformation information while preserving original evidence, generated Cyber DNA versions, and historical analytical results.

[00017] These and other aspects of the embodiments herein will be better appreciated and understood when considered in conjunction with the following description and the accompanying drawings. It should be understood, however, that the following descriptions, while indicating preferred embodiments and numerous specific details thereof, are given by way of illustration and not of limitation. Many changes and modifications may be made within the scope of the embodiments herein without departing from the spirit thereof, and the embodiments herein include all such modifications.

## BRIEF DESCRIPTION OF THE DRAWINGS

[00018] The embodiments herein will be better understood from the following detailed description with reference to the drawings, in which:

[00019] FIG. 1 illustrates a high-level environment for dynamic Cyber DNA fingerprinting and evolutionary analysis of cybercrime complaints according to one or more embodiments herein;

[00020] FIG. 2 illustrates a detailed architecture of a system for dynamic Cyber DNA fingerprinting and evolutionary analysis of cybercrime complaints according to one or more embodiments herein;

[00021] FIG. 3 illustrates an exemplary Dynamic Cyber DNA fingerprint comprising interconnected genes according to one or more embodiments herein;

[00022] FIG. 4 illustrates a cross-complaint comparison and mutation analysis of Dynamic Cyber DNA fingerprints according to one or more embodiments herein;

[00023] FIG. 5 illustrates an exemplary Cyber DNA evolutionary lineage and Crime Evolution Memory according to one or more embodiments herein;

[00024] FIG. 6 illustrates an exemplary candidate campaign profile and future Cyber DNA variant analysis according to one or more embodiments herein; and

[00025] FIG. 7 illustrates a flow diagram of a method for dynamic Cyber DNA fingerprinting and evolutionary analysis of cybercrime complaints according to one or more embodiments herein.

## DETAILED DESCRIPTION OF THE DRAWINGS

[00026] The embodiments herein and the various features and advantageous details thereof are explained more fully with reference to the non-limiting embodiments that are illustrated in the accompanying drawings and detailed in the following description. Descriptions of well-known components and processing techniques are omitted so as to not unnecessarily obscure the embodiments herein. The examples used herein are intended merely to facilitate an understanding of ways in which the embodiments herein may be practiced and to further enable those of skill in the art to practice the embodiments herein. Accordingly, the examples should not be construed as limiting the scope of the embodiments herein.

[00027] As mentioned, there remains a need for an improved system and method for analysing cybercrime complaints by generating dynamic behavioural fingerprints from multimodal evidence, identifying relationships between seemingly unrelated complaints, detecting changes and preserved behavioural characteristics, tracking evolutionary relationships among cybercrime patterns, and providing evidence-linked and explainable investigative intelligence. The embodiments described herein address these needs through Dynamic Cyber DNA fingerprinting, cross-complaint correlation, mutation analysis, evolutionary lineage analysis, campaign-level intelligence, and adaptive updating based on investigative feedback. Referring now to the drawings, and more particularly to FIGS. 1 to 7, where similar reference characters denote corresponding features consistently throughout the figures, there are shown preferred embodiments.

[00028] FIG. 1 illustrates a high-level operating environment 100 for dynamic Cyber DNA fingerprinting and evolutionary analysis of cybercrime complaints according to one or more embodiments herein. The operating environment 100 enables a complainant to submit a cybercrime complaint and associated digital evidence to a Dynamic Cyber DNA analysis system 102 and enables an authorized investigator to receive an evidence-linked investigation output and provide investigative feedback. The operating environment 100 may further include a

communication network 104, a data repository 106, a complainant computing device 108, an investigator computing device 110, and one or more external information sources 112. The Dynamic Cyber DNA analysis system 102 includes one or more processors 102A and a memory 102B for storing and executing instructions associated with the analysis of cybercrime complaints. The system 102 is configured to receive cybercrime complaints and associated digital evidence, process information associated with the complaints, generate Dynamic Cyber DNA fingerprints, identify relationships between cybercrime complaints, analyse changes between related cybercrime patterns, and provide investigation information to an authorized investigator.

[00029] The complainant computing device 108 may be any electronic computing device configured to enable a user or complainant to communicate with the Dynamic Cyber DNA analysis system 102. In some embodiments, the complainant computing device 108 may include a desktop computer, laptop computer, mobile phone, tablet computer, or another electronic device having communication capabilities. The complainant computing device 108 enables submission of a cybercrime complaint and associated digital evidence to the system 102. The cybercrime complaint may include, for example, a complaint narrative, date and time information associated with an incident, a suspected cybercrime category, phone numbers, email addresses, URLs, domain names, UPI identifiers, bank account details, transaction identifiers, financial loss information, and information identifying a communication platform used by a suspected attacker. The associated digital evidence may include one or more screenshots, images, SMS records, email records, chat records, transaction receipts, bank statements, QR codes, documents, audio recordings, or other supporting digital evidence.

[00030] In one example embodiment, a complainant using the computing device 108 may submit a complaint indicating that a person claiming to be a bank representative contacted the complainant through a communication platform and represented that a KYC verification was required to prevent blockage of a bank account. The complaint may further indicate that a link was provided, OTP information was obtained, and an unauthorized UPI transaction subsequently occurred. The complainant may additionally submit supporting evidence, such as a communication screenshot, an SMS containing OTP-related information, a suspicious URL, a transaction receipt, a bank statement, and a call recording. The system 102 may associate the

submitted information and evidence with a complaint identifier while preserving the original evidence for subsequent analysis and traceability.

[00031] The communication network 104 communicatively couples the complainant computing device 108, the Dynamic Cyber DNA analysis system 102, and the investigator computing device 110. The communication network 104 may facilitate transmission of cybercrime complaints, associated digital evidence, investigation information, and investigator feedback between the respective components of the operating environment 100. In some embodiments, the communication network 104 may include one or more wired networks, wireless networks, local networks, wide-area networks, or combinations thereof.

[00032] The Dynamic Cyber DNA analysis system 102 includes the processor 102A and the memory 102B. The processor 102A may execute instructions stored in the memory 102B to perform the operations associated with Dynamic Cyber DNA fingerprinting and evolutionary analysis. The memory 102B may store cybercrime complaint information, associated digital evidence, extracted and normalized information, generated Cyber DNA fingerprints, relationship information, mutation information, evolutionary relationship information, campaign information, and historical analytical information. The processor 102A may cooperate with the memory 102B to enable the Dynamic Cyber DNA fingerprints to be updated when additional evidence or validated investigative information becomes available while preserving historical information and evidence provenance.

[00033] In some embodiments, the processor 102A processes heterogeneous evidence received from the complainant computing device 108 by identifying respective evidence types and extracting relevant cybercrime information from the evidence. For example, complaint text may be processed to identify entities, intent-bearing statements, amounts, and attack actions; screenshots may be processed to identify messages, phone numbers, URLs, and UPI identifiers; communications may be processed to identify senders, message content, OTP requests, and URLs; and transaction-related documents may be processed to identify amounts, transaction identifiers, timestamps, and other transaction information. The extracted information may be normalized and assessed to determine whether information from multiple evidence sources supports a common extracted fact.

[00034] The processor 102A may further analyse the extracted information to identify behavioural intentions and an attack sequence associated with a reported cybercrime. In some

embodiments, the identified behavioural information is represented using interconnected Cyber DNA genes including an intent gene, a behaviour gene, a communication gene, a financial gene, a temporal gene, an infrastructure gene, and an evidence gene. Each Cyber DNA gene may represent a corresponding characteristic associated with the cybercrime complaint and may include a gene type, an encoded characteristic, an associated confidence value, a temporal position, and one or more relationships with other Cyber DNA genes. Accordingly, the generated Dynamic Cyber DNA fingerprint may preserve not only individual characteristics associated with a cybercrime, but also confidence, sequence, and relationships between the characteristics.

[00035] The data repository 106 is communicatively coupled to the Dynamic Cyber DNA analysis system 102 and may store information used during analysis of cybercrime complaints. In some embodiments, the data repository 106 stores complaint information, associated evidence, previously generated Cyber DNA fingerprints, candidate relationships, mutation information, evolutionary relationship information, and historical transformation information. The stored information enables the system 102 to compare a newly generated Dynamic Cyber DNA fingerprint with fingerprints associated with previously reported complaints and to identify potentially related incidents based on direct entity relationships and behavioural similarity. The system may thereby identify relationships even when observable characteristics of related cybercrime incidents differ.

[00036] In some embodiments, the system 102 determines changes between corresponding characteristics of related Dynamic Cyber DNA fingerprints while identifying behavioural characteristics that remain preserved. The determined changes may include characteristics that are modified, substituted, inserted, removed, or otherwise transformed. For example, a communication characteristic may change from SMS to WhatsApp, an attack method may change from OTP collection to a fake login mechanism or remote-access technique, an impersonation characteristic may change, or a payment mechanism may change from UPI to a digital wallet. Despite such changes, an underlying behavioural strategy may remain substantially preserved. The system may determine a Context Mutation Score and a Behavioural Core Preservation Score for assessing the relationship between related cybercrime patterns.

[00037] In some embodiments, the system 102 uses determined mutation relationships, temporal information, preserved behavioural characteristics, and supporting evidence to generate a candidate Cyber DNA evolutionary lineage. The evolutionary lineage represents candidate

relationships between Cyber DNA variants and may indicate how an observed cybercrime pattern changes or branches over time. For example, related Cyber DNA fingerprints associated with SMS-based, WhatsApp-based, and subsequently modified attack patterns may be arranged according to their temporal occurrence and identified mutation paths. Each candidate evolutionary relationship may be supported by temporal ordering, behavioural-core preservation, mutation characteristics, and supporting evidence, without independently establishing that the same offender or criminal organization performed the associated attacks.

[00038] The system 102 may further maintain historical transformation information associated with previously observed and sufficiently supported or investigator-validated changes between Cyber DNA characteristics. In some embodiments, such historical transformation information is used as Crime Evolution Memory to assist analysis of subsequently received complaints. The system 102 may group related Cyber DNA fingerprints into a candidate campaign profile representing behavioural characteristics that remain stable and operational characteristics that change across a related set of cybercrime complaints. Based on previously observed transformation patterns and relevant contextual information, the system may further estimate one or more plausible future variants of a Cyber DNA characteristic. Each candidate future variant may be associated with a transition probability or confidence value and supporting historical transition information.

[00039] The investigator computing device 110 is configured to enable an authorized investigator to receive and review investigation information generated by the Dynamic Cyber DNA analysis system 102. In some embodiments, the investigation information includes information identifying related complaints, generated Cyber DNA characteristics, detected changes, preserved behavioural characteristics, candidate evolutionary relationships, supporting evidence, and confidence information. The investigation output may retain references to supporting evidence and intermediate analytical information, thereby enabling the investigator to assess why complaints were associated and how a Cyber DNA pattern was determined to have changed.

[00040] In some embodiments, the investigator computing device 110 enables the investigator to provide feedback indicating that a generated relationship, mutation relationship, or campaign association is confirmed, rejected, or requires further investigation. Confirmed information may be retained as validated relationship information, rejected information may be

retained as negative validation information, and information requiring further investigation may remain unresolved until additional evidence or assessment becomes available. The feedback may be used by the system 102 to update or recalibrate entity-reliability weights, Cyber DNA similarity parameters, mutation relationships, evolutionary lineage assessment parameters, and historical transformation information, while preserving original evidence, generated Cyber DNA versions, and historical analytical results.

[00041] The one or more external information sources 112 may be communicatively coupled to the Dynamic Cyber DNA analysis system 102 to provide additional information for use in analysing a cybercrime complaint. In some embodiments, information received from the external information sources 112 may be used together with complaint information, associated digital evidence, previously generated Cyber DNA fingerprints, and historical relationship information maintained by the system 102. The external information sources 112 may therefore provide supplementary information while the system 102 maintains the complaint-level Cyber DNA analysis and historical analytical information used for identifying relationships and evolutionary patterns.

[00042] Thus, the operating environment 100 provides an integrated environment in which the complainant computing device 108 communicates a cybercrime complaint and associated digital evidence through the communication network 104 to the Dynamic Cyber DNA analysis system 102, the system 102 processes and analyses the received information using the processor 102A and memory 102B, accesses or stores relevant information in the data repository 106, optionally utilizes information from the one or more external information sources 112, and provides an evidence-linked investigation output to the investigator computing device 110. The investigator may further provide feedback through the operating environment 100 to improve subsequent analysis of evolving cybercrime patterns.

[00043] FIG. 2 illustrates an internal functional architecture of the Dynamic Cyber DNA analysis system 102 according to one or more embodiments herein. The Dynamic Cyber DNA analysis system 102 includes a complaint and digital evidence processing module 202, an entity and cybercrime event extraction and normalization module 204, a behavioural intent and attack sequence determination module 206, a Dynamic Cyber DNA fingerprint generation module 208, a cross-complaint relationship identification module 210, a change and preserved behavioural characteristic determination module 212, a Cyber DNA evolutionary lineage generation module

214, and an evidence-linked investigation output generation module 216. The modules are communicatively coupled to enable sequential processing of cybercrime complaint information and generation of investigation information.

[00044] The complaint and digital evidence processing module 202 receives cybercrime complaint information and associated digital evidence for subsequent analysis. The complaint and digital evidence processing module 202 provides the received information to the entity and cybercrime event extraction and normalization module 204.

[00045] The entity and cybercrime event extraction and normalization module 204 extracts entities and cybercrime events from the received information, normalizes the extracted entities and cybercrime events, and determines evidence support associated therewith. The normalized entities, cybercrime events, and evidence support information are provided to the behavioural intent and attack sequence determination module 206.

[00046] The behavioural intent and attack sequence determination module 206 determines behavioural intents and an attack sequence associated with the cybercrime events. In some embodiments, the behavioural intent and attack sequence determination module 206 identifies respective micro-intents associated with the cybercrime events and arranges the respective micro-intents according to the attack sequence. Each micro-intent may be associated with corresponding supporting evidence and a confidence value based on semantic correspondence, evidence support, and contextual consistency.

[00047] The Dynamic Cyber DNA fingerprint generation module 208 generates a Dynamic Cyber DNA fingerprint based on the extracted and normalized entities, cybercrime events, behavioural intents, attack sequence, evidence support, and temporal information. The Dynamic Cyber DNA fingerprint includes interconnected genes representing intent characteristics, behavioural characteristics, communication characteristics, financial characteristics, temporal characteristics, infrastructure characteristics, and evidence characteristics.

[00048] In some embodiments, each gene includes a gene type, an encoded characteristic, an associated confidence value, a temporal position, and one or more relationships with other genes. Accordingly, the Dynamic Cyber DNA fingerprint preserves characteristics associated with a cybercrime complaint together with a sequence and interrelationship of the characteristics.

[00049] The cross-complaint relationship identification module 210 receives the generated Dynamic Cyber DNA fingerprint and compares the generated Dynamic Cyber DNA fingerprint with previously generated Cyber DNA fingerprints 218. The cross-complaint relationship identification module 210 identifies candidate relationships between cybercrime complaints based on direct entity relationships and behavioural similarity.

[00050] In some embodiments, the cross-complaint relationship identification module 210 determines an entity relationship score based on shared entities and an adaptive reliability weight associated with each entity type, determines a Cyber DNA similarity score based on similarities between corresponding genes, and combines the entity relationship score and the Cyber DNA similarity score to generate a hybrid relationship score.

[00051] The change and preserved behavioural characteristic determination module 212 receives information associated with a candidate relationship and determines changes between corresponding characteristics of the Dynamic Cyber DNA fingerprints. The change and preserved behavioural characteristic determination module 212 further identifies behavioural characteristics that are preserved despite the determined changes.

[00052] In some embodiments, the change and preserved behavioural characteristic determination module 212 identifies characteristics that are preserved, modified, substituted, inserted, removed, or transformed between the Dynamic Cyber DNA fingerprints. The change and preserved behavioural characteristic determination module 212 further determines a behavioural core preservation measure and a context mutation measure for distinguishing a candidate evolved variant from a merely similar cybercrime complaint.

[00053] The Cyber DNA evolutionary lineage generation module 214 generates a candidate Cyber DNA evolutionary lineage based on the determined changes, the preserved behavioural characteristics, temporal information, and evidence supporting the candidate relationship. The Cyber DNA evolutionary lineage generation module 214 thereby represents candidate evolutionary relationships between related Cyber DNA fingerprints.

[00054] The evidence-linked investigation output generation module 216 generates an investigation output identifying the candidate relationship, the determined changes, the preserved behavioural characteristics, and the candidate Cyber DNA evolutionary lineage. The investigation output may further provide evidence information supporting the generated analytical results.

[00055] In some embodiments, the memory module 224 stores information used by one or more of the modules 210, 212, 214, and 216. The memory module 224 may store previously generated Cyber DNA fingerprints, historical cybercrime relationship information, mutation information, and other historical analytical information. The stored information may be used to support comparison, mutation analysis, evolutionary lineage generation, and generation of the evidence-linked investigation output.

[00056] FIG. 3 illustrates an exemplary Dynamic Cyber DNA fingerprint 300 comprising a plurality of interconnected Cyber DNA genes according to one or more embodiments herein. The Dynamic Cyber DNA fingerprint 300 provides a structured representation of behavioural and operational characteristics associated with a cybercrime complaint. Unlike a representation based only on a crime category, keywords, or individual identifiers, the Dynamic Cyber DNA fingerprint 300 represents an underlying behavioural strategy together with observable characteristics of an attack. The Dynamic Cyber DNA fingerprint 300 may include an intent gene 302, a behaviour gene 304, a communication gene 306, a financial gene 308, a temporal gene 310, an infrastructure gene 312, and an evidence gene 314. The respective Cyber DNA genes may be interconnected to preserve relationships, sequence, and contextual associations between characteristics associated with the cybercrime complaint.

[00057] At 302, the intent gene 302 represents behavioural intentions associated with an attack. The intent gene 302 may represent a sequence of intentions or behavioural objectives identified from information associated with the cybercrime complaint. In one example embodiment involving an authority impersonation-based cybercrime, the intent gene 302 may represent authority impersonation, establishment of legitimacy, creation of fear and urgency, credential acquisition, and financial extraction. In another example involving a recruitment-related fraud, the intent gene 302 may represent trust establishment, opportunity attraction, creation of urgency, financial commitment, and repeated financial extraction. Thus, the intent gene 302 may represent how an attacker attempts to influence a victim during progression of the attack.

[00058] At 304, the behaviour gene 304 represents a sequence of actions used to execute the cybercrime attack. In one example embodiment, the behaviour gene 304 may represent a fake KYC process, delivery of a malicious link, collection of OTP information, and an unauthorized transfer. In another example involving recruitment-related fraud, the behaviour gene 304 may

represent a fake recruitment process, a job offer, collection of a registration fee, and collection of a security deposit. The behaviour gene 304 thereby represents an operational progression of attack actions associated with the reported cybercrime.

[00059] At 306, the communication gene 306 represents one or more communication channels and interaction methods used during the cybercrime attack. For example, the communication gene 306 may identify WhatsApp, Telegram, SMS, email, a chat platform, or another communication channel through which an attacker interacts with a victim. The communication gene 306 may enable a communication characteristic to be represented separately from an underlying behavioural strategy, such that a change in a communication channel may subsequently be identified while other behavioural characteristics remain preserved.

[00060] At 308, the financial gene 308 represents a payment method, transaction pattern, financial amount, or other financial characteristic associated with the cybercrime complaint. For example, the financial gene 308 may represent a UPI-based transaction involving an amount of ₹48,000 in an unauthorized transfer scenario. In another example, the financial gene 308 may represent a first UPI-based payment of ₹5,000 and a subsequent payment of ₹18,000, resulting in a total amount of ₹23,000 in a two-stage recruitment-related fraud. The financial gene 308 may thereby preserve one or more relationships between payment characteristics and other characteristics represented by the Dynamic Cyber DNA fingerprint 300.

[00061] At 310, the temporal gene 310 represents timing, duration, temporal ordering, or another temporal characteristic associated with attack events. For example, the temporal gene 310 may represent a short-duration attack followed by an evening transaction, or a two-stage payment occurring within twenty-four hours. The temporal gene 310 may therefore preserve the ordering and timing context of characteristics represented by the other Cyber DNA genes.

[00062] At 312, the infrastructure gene 312 represents digital infrastructure associated with the cybercrime attack. The infrastructure gene 312 may include one or more phone numbers, URLs, domain names, UPI identifiers, accounts, or other digital infrastructure indicators. For example, the infrastructure gene 312 may represent an association among a phone number, a suspicious URL, and a UPI ID. In another example, the infrastructure gene 312 may represent a Telegram account associated with a UPI ID.

[00063] At 314, the evidence gene 314 represents evidence supporting one or more characteristics represented by the other Cyber DNA genes. The evidence gene 314 may include a

complaint record, screenshots, transaction receipts, bank statements, messages, documents, or other associated digital evidence. For example, the evidence gene 314 may represent an evidentiary association among a complaint, a screenshot, a transaction receipt, and a bank statement. In another example, the evidence gene 314 may represent screenshots, a job offer PDF, UPI receipts, and a bank statement supporting the corresponding cybercrime complaint. The evidence gene 314 enables the Dynamic Cyber DNA fingerprint 300 to retain an association between represented characteristics and corresponding supporting evidence.

[00064] At 316, an individual Cyber DNA gene may be represented as a structured gene record. The individual Cyber DNA gene 316 may include a gene type 318, an encoded characteristic 320, a confidence value 322, a temporal position 324, and gene relationship information 326. Accordingly, the individual Cyber DNA gene 316 is represented not merely as a text value, but as a structured representation that preserves the corresponding characteristic and information associated with the characteristic.

[00065] At 318, the gene type 318 identifies a category of the individual Cyber DNA gene 316. The gene type 318 may identify the individual Cyber DNA gene 316 as an intent gene, behaviour gene, communication gene, financial gene, temporal gene, infrastructure gene, or evidence gene.

[00066] At 320, the encoded characteristic 320 represents a value or characteristic associated with the gene type 318. For example, when the individual Cyber DNA gene 316 is a behaviour gene, the encoded characteristic 320 may represent OTP collection, whereas an encoded characteristic associated with a communication gene may represent a communication channel used during the cybercrime attack.

[00067] At 322, the confidence value 322 represents a confidence associated with the individual Cyber DNA gene 316. The confidence value 322 may indicate a strength of support for the corresponding gene based on information obtained from one or more sources. In some embodiments, the confidence value may be determined based on behavioural confidence, evidence support, and relationship consistency with other Cyber DNA genes.

[00068] At 324, the temporal position 324 represents a position of the individual Cyber DNA gene 316 within an attack sequence or another temporal context associated with the cybercrime complaint. For example, an OTP-related behaviour gene may have a temporal

position indicating occurrence after delivery of a malicious link and before an unauthorized financial transaction.

[00069] At 326, the gene relationship information 326 represents one or more relationships between the individual Cyber DNA gene 316 and other Cyber DNA genes. For example, an OTP-related behaviour gene may be associated with an intent relating to credential acquisition and a financial characteristic associated with a subsequent unauthorized transaction. The gene relationship information 326, together with the temporal position 324, enables the Dynamic Cyber DNA fingerprint 300 to preserve not only which characteristics are present, but also their confidence, sequence, and relationships with other characteristics.

[00070] In some embodiments, the respective genes 302, 304, 306, 308, 310, 312, and 314 collectively form a Dynamic Cyber DNA fingerprint associated with a corresponding cybercrime complaint. The interconnected representation enables underlying behavioural characteristics and observable operational characteristics to be retained in a structured form and enables the Dynamic Cyber DNA fingerprint to be incrementally updated when new evidence or related complaints become available. The resulting representation may subsequently support comparison of Cyber DNA fingerprints, identification of potentially related complaints, and analysis of changes in cybercrime patterns while identifying characteristics that remain preserved.

[00071] FIG. 4 illustrates an exemplary cross-complaint Dynamic Cyber DNA comparison and mutation analysis 400 for identifying preserved behavioural characteristics and changed contextual characteristics according to one or more embodiments herein. The cross-complaint mutation analysis 400 compares a first Dynamic Cyber DNA fingerprint 402 and a second Dynamic Cyber DNA fingerprint 404 associated with a candidate relationship. The analysis enables the system to determine how one Cyber DNA fingerprint differs from another while identifying fundamental behavioural characteristics that remain substantially unchanged. Accordingly, the analysis assists in distinguishing between merely similar cybercrime complaints and a possible evolved variant of an existing cybercrime pattern.

[00072] At 402, the first Dynamic Cyber DNA fingerprint 402 represents characteristics associated with a first or earlier cybercrime complaint. The first Dynamic Cyber DNA fingerprint 402 may include behavioural, contextual, communication, impersonation, attack-method, financial, temporal, infrastructure, and evidence characteristics associated with the first complaint.

[00073] At 404, the second Dynamic Cyber DNA fingerprint 404 represents characteristics associated with a second cybercrime complaint identified as having a candidate relationship with the first cybercrime complaint. The second Dynamic Cyber DNA fingerprint 404 may represent a later observed variant having one or more characteristics that differ from corresponding characteristics represented by the first Dynamic Cyber DNA fingerprint 402.

[00074] At 406, the first Dynamic Cyber DNA fingerprint 402 and the second Dynamic Cyber DNA fingerprint 404 are compared to identify corresponding Cyber DNA characteristics and determine whether the characteristics are preserved or have undergone a contextual mutation. The comparison may identify characteristics that are preserved, modified, substituted, inserted, removed, or transformed. The system therefore does not treat every difference between the two cybercrime complaints as evidence that the complaints are unrelated. Instead, the system determines whether the identified differences represent contextual mutations around a preserved behavioural core.

[00075] In one example embodiment, the first Dynamic Cyber DNA fingerprint 402 may represent a cybercrime complaint involving impersonation of a bank officer, communication through SMS, OTP collection as an attack method, and UPI as a payment channel. The second Dynamic Cyber DNA fingerprint 404 may represent a related cybercrime complaint involving impersonation of an RBI officer, communication through WhatsApp, use of a remote-access application, and a digital wallet as a payment channel. Although the observable operational characteristics differ, the complaints may preserve a common underlying behavioural strategy involving authority impersonation, induction of fear, creation of urgency, victim control, and financial extraction.

[00076] At 408, the analysis identifies preserved behavioural characteristics that remain substantially common between the first Dynamic Cyber DNA fingerprint 402 and the second Dynamic Cyber DNA fingerprint 404. The preserved behavioural characteristics may form a behavioural core associated with the candidate relationship. In the example embodiment, a psychological strategy involving fear and urgency and an objective involving financial extraction may be identified as preserved characteristics despite changes in impersonation, communication, attack method, and payment channel.

[00077] At 410, the analysis identifies modified characteristics. A modified characteristic may represent a corresponding Cyber DNA characteristic that remains within a generally similar

category but has a different value or operational context in the second Dynamic Cyber DNA fingerprint 404. For example, impersonation of a bank officer may be modified to impersonation of an RBI officer, communication through SMS may be modified to communication through WhatsApp, an OTP collection method may be modified to a remote-access application, or a UPI-based payment channel may be modified to a digital-wallet-based payment channel.

[00078] At 412, the analysis identifies substituted characteristics. A substituted characteristic may represent replacement of a characteristic or attack mechanism associated with the first Dynamic Cyber DNA fingerprint 402 by a different characteristic or attack mechanism associated with the second Dynamic Cyber DNA fingerprint 404. For example, a communication mechanism, credential-acquisition mechanism, impersonation characteristic, or payment mechanism associated with an earlier attack pattern may be replaced by a corresponding but different mechanism in a later attack pattern. Such substitution may represent an operational change while the underlying behavioural objective remains preserved.

[00079] At 414, the analysis identifies inserted characteristics. An inserted characteristic may represent a characteristic present in the second Dynamic Cyber DNA fingerprint 404 that is not present in the first Dynamic Cyber DNA fingerprint 402. For example, a newly observed operational step, communication characteristic, infrastructure characteristic, or attack mechanism may be inserted into the attack pattern represented by the later fingerprint.

[00080] At 416, the analysis identifies removed characteristics. A removed characteristic may represent a characteristic present in the first Dynamic Cyber DNA fingerprint 402 that is absent from the second Dynamic Cyber DNA fingerprint 404. The removal of a characteristic may be considered together with other preserved and changed characteristics when determining whether the second cybercrime complaint represents a possible evolved variant rather than an unrelated complaint.

[00081] At 418, the analysis identifies transformed characteristics. A transformed characteristic may represent a characteristic that has changed from one operational form in the first Dynamic Cyber DNA fingerprint 402 to another operational form in the second Dynamic Cyber DNA fingerprint 404. For example, historical transformation patterns may include changes from SMS-based communication to WhatsApp-based communication, from OTP-related credential acquisition to remote-access techniques, or from UPI-based payments to other

financial channels. The identified transformation may subsequently be represented as part of a mutation path describing how an earlier Cyber DNA pattern could correspond to a later variant.

[00082] At 420, a behavioural core preservation measure is determined for assessing an extent to which the fundamental behavioural strategy remains preserved between the first Dynamic Cyber DNA fingerprint 402 and the second Dynamic Cyber DNA fingerprint 404. The measure may be determined based on similarities between respective behavioural components and importance values assigned to the behavioural components. In one illustrative example, authority impersonation may have a similarity of 0.9 and a weight of 0.2, fear induction may have a similarity of 1.0 and a weight of 0.2, urgency creation may have a similarity of 1.0 and a weight of 0.2, victim control may have a similarity of 0.8 and a weight of 0.15, and financial extraction may have a similarity of 1.0 and a weight of 0.25. Based on these illustrative values, the behavioural core preservation measure may indicate preservation of approximately 95 percent of the evaluated behavioural core. The numerical values are illustrative, and the importance values may be calibrated using validated historical cases.

[00083] At 422, a context mutation measure is determined for assessing an extent to which the operational context has changed between the first Dynamic Cyber DNA fingerprint 402 and the second Dynamic Cyber DNA fingerprint 404. The context mutation measure may be based on differences between selected contextual components and corresponding importance values. In one illustrative example, a change from SMS to WhatsApp may have a difference value of 0.8, a change in the impersonated entity from a bank to RBI may have a difference value of 0.4, a change in the credential method from OTP to remote access may have a difference value of 0.9, and a change in the payment channel from UPI to a wallet may have a difference value of 0.7. When equal importance values are used for illustration, the context mutation measure may be 0.70, indicating a 70 percent contextual mutation across the evaluated operational characteristics. The context mutation measure does not indicate that the cybercrime complaints are 70 percent different overall, but instead indicates the degree of change in the contextual components selected for mutation analysis.

[00084] Following identification of preserved and changed characteristics, a mutation path may be generated to describe how the first Dynamic Cyber DNA fingerprint 402 could correspond to the second Dynamic Cyber DNA fingerprint 404. The mutation path provides an

explainable representation of what changed and what remained stable between the cybercrime patterns, rather than providing only an overall similarity value.

[00085] At 424, a candidate evolved variant assessment is generated based on the candidate relationship, the behavioural core preservation measure, and the context mutation measure. In some embodiments, the assessment further considers a hybrid relationship score associated with the candidate relationship. A combined mutation assessment may use calibrated importance values associated with the hybrid relationship score, the behavioural core preservation measure, and the context mutation measure. A high context mutation measure does not automatically reject the candidate relationship. For example, a high degree of operational change together with a high degree of behavioural core preservation may indicate that the operational method has changed substantially while the underlying behavioural strategy remains stable. The individual analytical values may therefore be retained along with the combined assessment for investigator interpretation.

[00086] In some embodiments, the results of the cross-complaint mutation analysis 400 are recorded in a Cyber DNA Mutation Record. The Cyber DNA Mutation Record may include the candidate Cyber DNA pair, the preserved behavioural core, modified genes, the context mutation measure, the behavioural core preservation measure, the mutation path, and the mutation assessment. The resulting candidate mutation relationship may subsequently be provided for Cyber DNA evolutionary lineage analysis to determine how sufficiently supported mutations can be organized across multiple cybercrime complaints and over time.

[00087] FIG. 5 illustrates an exemplary Cyber DNA evolutionary lineage and Crime Evolution Memory according to one or more embodiments herein. The FIG. 5 illustrates how sufficiently supported Cyber DNA variants associated with related cybercrime complaints may be organized into a temporal evolutionary lineage and how observed transformations may be retained in a Crime Evolution Memory. Unlike a flat grouping of related complaints, the Cyber DNA evolutionary lineage represents how an attack pattern may change from one Cyber DNA variant to another while retaining significant portions of an underlying behavioural core. The Crime Evolution Memory stores previously observed and sufficiently supported or investigator-validated Cyber DNA transformations so that subsequent complaints may be evaluated in view of known evolutionary patterns rather than only as isolated incidents.

[00088] At 502, a first Cyber DNA variant 502 represents a Cyber DNA fingerprint associated with an earlier complaint or an earlier observed stage of a candidate cybercrime evolution. The first Cyber DNA variant 502 may include communication, impersonation, attack-method, payment, behavioural, temporal, infrastructure, and other characteristics represented by the corresponding Dynamic Cyber DNA fingerprint.

[00089] In one example embodiment, the first Cyber DNA variant 502 may be associated with communication through SMS, impersonation of a bank officer, OTP collection as an attack method, and UPI as a payment channel. The first Cyber DNA variant 502 may therefore represent an earlier observed operational configuration of a cybercrime pattern.

[00090] At 504, a second Cyber DNA variant 504 represents a Cyber DNA fingerprint associated with a subsequent complaint having a candidate evolutionary relationship with the first Cyber DNA variant 502. The second Cyber DNA variant 504 may preserve one or more characteristics associated with the first Cyber DNA variant 502 while incorporating one or more changes identified through the cross-complaint mutation analysis.

[00091] In one example embodiment, the second Cyber DNA variant 504 may be associated with communication through WhatsApp, impersonation of a bank officer, use of a fake login page as an attack method, and UPI as a payment channel. Thus, the second Cyber DNA variant 504 may retain certain characteristics associated with the first Cyber DNA variant 502 while representing changes in communication and attack method.

[00092] At 506, a third Cyber DNA variant 506 represents a further Cyber DNA fingerprint associated with another complaint having a candidate evolutionary relationship with the preceding Cyber DNA variants. The third Cyber DNA variant 506 may represent a further evolution of operational characteristics while retaining significant portions of the behavioural core.

[00093] In one example embodiment, the third Cyber DNA variant 506 may be associated with communication through WhatsApp, impersonation of an RBI officer, use of a remote-access technique as an attack method, and use of a digital wallet as a payment channel. Accordingly, the sequence of variants may demonstrate changes in communication, impersonation, attack method, and payment characteristics across related complaints.

[00094] At 508, a first candidate evolutionary relationship 508 represents a candidate relationship between the first Cyber DNA variant 502 and the second Cyber DNA variant 504.

The first candidate evolutionary relationship 508 may be determined based on temporal ordering, preserved behavioural characteristics, mutation characteristics, and supporting evidence. The relationship represents a candidate evolution of an attack pattern and does not, by itself, establish that the same offender or criminal organization performed the attacks associated with the respective complaints.

[00095] At 510, a second candidate evolutionary relationship 510 represents a candidate relationship between the second Cyber DNA variant 504 and the third Cyber DNA variant 506. The second candidate evolutionary relationship 510 may likewise be based on the temporal occurrence of the respective complaints, preservation of behavioural characteristics, identified mutation characteristics, and supporting evidence. The candidate evolutionary relationships may collectively organize the Cyber DNA variants 502, 504, and 506 into a temporal evolutionary sequence representing a possible progression of the cybercrime pattern.

[00096] At 512, a preserved behavioural core 512 represents behavioural characteristics that remain substantially preserved across the Cyber DNA variants 502, 504, and 506 despite changes in their observable operational characteristics. In some embodiments, the preserved behavioural core 512 may include authority-based influence, induction of fear, creation of urgency, victim control, and financial extraction. Accordingly, FIG. 5 illustrates that communication channels, impersonated entities, attack methods, and payment mechanisms may change over time while the underlying behavioural strategy remains substantially preserved.

[00097] At 514, a lineage confidence assessment 514 represents an assessment of the confidence associated with a candidate evolutionary relationship before the relationship is retained in the Cyber DNA evolutionary lineage. The lineage confidence assessment 514 may be based on structural similarity between corresponding Cyber DNA variants, mutation cost associated with transforming one variant into another, behavioural core preservation, temporal plausibility of one variant occurring before another, and evidence support for the relationship. The respective factors may be associated with calibrated importance values and may be normalized to produce a lineage confidence value.

[00098] The mutation cost may reduce the lineage confidence when a proposed transformation is excessively large or implausible. Conversely, high structural similarity, strong behavioural preservation, a reasonable mutation cost, valid temporal ordering, and sufficient evidence support may result in a higher lineage confidence. A high lineage confidence indicates

a strong candidate evolutionary relationship and does not constitute proof of common criminal attribution.

[00099] At 516, the Crime Evolution Memory 516 retains information associated with observed Cyber DNA transformations. Once candidate mutation relationships are sufficiently supported or investigator-validated, corresponding observed transformations may be stored in the Crime Evolution Memory 516. The Crime Evolution Memory 516 thereby maintains information identifying which Cyber DNA characteristics have previously changed into other characteristics and the contexts in which the transformations occurred. The stored information can subsequently support evaluation of new Cyber DNA mutation relationships and subsequent analysis of evolving cybercrime patterns.

[000100] At 518, observed transformation records 518 represent historical records of transformations between Cyber DNA characteristics. The observed transformation records 518 may identify changes in communication methods, impersonation characteristics, attack methods, payment mechanisms, or other Cyber DNA characteristics. For example, historical transformation records may indicate that an OTP-collection characteristic has, in different validated cases, changed to a remote-access technique, a fake-login technique, or a malicious APK-based technique. The respective observed alternatives may be retained as separate historical transformation records.

[000101] At 520, contextual transition information 520 represents contextual information associated with the observed transformation records 518. The contextual transition information 520 enables the Crime Evolution Memory 516 to preserve not only the fact that one characteristic has changed to another characteristic, but also the relevant cybercrime context in which the transformation was observed. In some embodiments, the contextual transition information 520 may support determination of a contextual transition probability based on validated occurrences of a particular transformation within a relevant context and occurrences of alternative transformations from a corresponding characteristic.

[000102] For example, where validated historical cases indicate that OTP collection changed to remote access in eighteen cases, to a fake-login technique in ten cases, and to a malicious APK in seven cases, the Crime Evolution Memory 516 may retain information reflecting the relative occurrence of these observed transformations. Thus, the Crime Evolution Memory 516 can indicate that a remote-access technique has been observed more frequently as a

successor to OTP-based victim control than the other recorded alternatives within the evaluated context.

[000103] At 522, validated mutation information 522 represents mutation information that has been sufficiently supported or validated through investigator assessment. The validated mutation information 522 may include validated transformation relationships, associated behavioural characteristics, mutation paths, evidence support, and contextual information. Such information may be retained in the Crime Evolution Memory 516 for use in subsequent Cyber DNA analysis and evaluation of candidate evolutionary relationships.

[000104] At 524, subsequent Cyber DNA analysis 524 represents a subsequent analysis in which information retained in the Crime Evolution Memory 516 may be used when evaluating later-generated Dynamic Cyber DNA fingerprints. The stored evolutionary information may enable subsequent analysis to consider previously observed and validated transformations when determining candidate mutation relationships and broader evolutionary patterns. The generated Cyber DNA evolutionary lineage and Crime Evolution Memory may further support campaign-level analysis of groups of related Cyber DNA variants and estimation of plausible subsequent changes in operational characteristics.

[000105] In some embodiments, the original complaint evidence, generated Cyber DNA variants, historical analytical results, and information associated with evolutionary relationships may be retained to support traceability and subsequent analysis. The Cyber DNA evolutionary lineage and Crime Evolution Memory therefore provide a structured historical representation of how related cybercrime patterns may evolve over time while preserving information supporting the identified relationships and transformations.

[000106] FIG. 6 illustrates an exemplary candidate campaign profile and future Cyber DNA variant analysis 600 according to one or more embodiments herein. The candidate campaign profile and future Cyber DNA variant analysis 600 analyzes groups of related Cyber DNA variants and corresponding evolutionary information to identify a candidate cybercrime campaign and estimate plausible future changes in operational characteristics. The analysis is based on previously observed Cyber DNA mutations and does not predict a specific future cybercrime with certainty. Instead, the analysis identifies plausible attack-pattern variants based on historical and validated transition information.

[000107] At 602, a Cyber DNA evolutionary lineage 602 represents evolutionary relationships among related Cyber DNA variants associated with multiple cybercrime complaints. The Cyber DNA evolutionary lineage 602 provides information regarding the temporal progression of the related variants and changes identified between corresponding Cyber DNA characteristics. The evolutionary relationships may be sufficiently supported based on structural similarity, mutation characteristics, behavioural core preservation, temporal plausibility, and evidence support. The Cyber DNA evolutionary lineage 602 provides a basis for analyzing multiple related complaints collectively at a broader campaign level.

[000108] At 604, a Crime Evolution Memory 604 represents information associated with previously observed and sufficiently supported or investigator-validated Cyber DNA transformations. The Crime Evolution Memory 604 retains information identifying which Cyber DNA characteristics have previously transformed into other characteristics and the context in which the transformations occurred. Such historical transformation information may be used during subsequent analysis of related Cyber DNA variants and for estimation of plausible future changes.

[000109] At 606, a candidate campaign profile 606 is generated by grouping Cyber DNA fingerprints belonging to a sufficiently supported Cyber DNA evolutionary lineage. The candidate campaign profile 606 represents collective behavioural and evolutionary characteristics associated with related cybercrime complaints, rather than treating each complaint independently. The candidate campaign profile 606 may summarize characteristics that remain stable across the evolutionary lineage and characteristics that change as the cybercrime pattern evolves.

[000110] At 608, stable behavioural characteristics 608 represent behavioural characteristics that remain substantially preserved across the related Cyber DNA variants forming the candidate campaign profile 606. The stable behavioural characteristics 608 may represent an underlying behavioural strategy that persists despite changes in observable operational characteristics. For example, related variants may preserve behavioural characteristics associated with establishing authority or legitimacy, inducing fear or urgency, controlling or influencing a victim, acquiring credentials or other information, or achieving financial extraction, while one or more operational characteristics change over time. The identification of the stable behavioural characteristics 608 enables the candidate campaign

profile 606 to represent the underlying behavioural continuity of the related cybercrime complaints.

[000111] At 610, changing operational characteristics 610 represent Cyber DNA characteristics that differ across the related variants in the candidate campaign profile 606. The changing operational characteristics 610 may include changes in communication channels and interaction methods, impersonation characteristics, attack techniques, financial methods, infrastructure, or other observable operational characteristics. For example, a cybercrime pattern may evolve through changes in communication methods or attack mechanisms while retaining the stable behavioural characteristics 608. The changing operational characteristics 610 thereby identify how the operational implementation of the cybercrime pattern has changed across the evolutionary lineage.

[000112] At 612, a candidate future Cyber DNA variant 612 is estimated based on the transition patterns maintained in the Crime Evolution Memory 604 and the relevant characteristics associated with the candidate campaign profile 606. The estimation may involve identifying a current state of a selected Cyber DNA characteristic and determining a plausible subsequent state based on historical transformations observed within a relevant cybercrime context. A candidate state having a comparatively stronger contextual transition probability may be identified as a plausible future characteristic of the candidate future Cyber DNA variant 612.

[000113] In one example embodiment, where a current attack-method characteristic is represented as Fake Login and historical transition information indicates that a transition from Fake Login to Remote Access has been observed with a contextual transition probability of 0.68 within similar impersonation-based campaigns, the candidate future Cyber DNA variant 612 may include Remote Access as a plausible subsequent attack-method characteristic with an associated confidence value of 0.68. The historical transition and the corresponding campaign context may be retained to provide a basis for the estimation. The example is illustrative and does not indicate that such a future cybercrime will necessarily occur.

[000114] At 614, transition probability or confidence information 614 represents a contextual transition probability or confidence value associated with the candidate future Cyber DNA variant 612. The transition probability or confidence information 614 may be determined based on validated occurrences of a transformation from a current Cyber DNA characteristic to a candidate subsequent characteristic within a relevant context, relative to occurrences of

alternative observed transformations. In some embodiments, the determination may further incorporate a smoothing parameter and a number of possible observed transition categories.

[000115] For example, historical validated cases may indicate that an OTP Collection characteristic changed to Remote Access in eighteen cases, to Fake Login in ten cases, and to a Malicious APK in seven cases. Based on such historical information, the transition probability or confidence information 614 may indicate that Remote Access has been observed more frequently as a successor to OTP-based victim control than the other recorded alternatives within the evaluated context. This information may be used to support subsequent evaluation and estimation of plausible Cyber DNA transformations.

[000116] At 616, supporting historical transition information 616 represents one or more previously observed and validated Cyber DNA transformation patterns supporting the estimation of the candidate future Cyber DNA variant 612. The supporting historical transition information 616 may identify a current Cyber DNA characteristic, one or more observed subsequent characteristics, occurrences associated with the respective transformations, and corresponding contextual information. The supporting historical transition information 616 is retained with the estimated future variant to provide an explainable basis for identifying the candidate future characteristic.

[000117] At 618, relevant campaign context 618 represents contextual information associated with the candidate campaign profile 606 and the historical transformation information used for future variant estimation. The relevant campaign context 618 enables a transformation to be evaluated with reference to the circumstances in which similar transformations were previously observed, rather than treating all historical transformations as equally applicable to every cybercrime complaint. The relevant campaign context 618 may therefore be used in determining the contextual transition probability or confidence associated with the candidate future Cyber DNA variant 612.

[000118] At 620, a future variant intelligence record 620 is generated. The future variant intelligence record 620 may include the identified candidate campaign, one or more plausible future Cyber DNA variants, corresponding transition probability or confidence information, supporting historical transition information, and relevant contextual information. The future variant intelligence record 620 provides campaign-level intelligence regarding plausible

evolutionary changes in the cybercrime pattern and may subsequently be provided for investigator-facing explanation and validation.

[000119] In some embodiments, the candidate future Cyber DNA variant 612 is maintained as a plausible analytical estimation rather than a confirmed future event. The future variant intelligence record 620 therefore preserves the basis of the estimation, including the corresponding historical transformation pattern and contextual information, so that an investigator can evaluate the reasoning supporting the identified candidate future variant.

[000120] FIG. 7 illustrates a flow diagram of a method for analyzing cybercrime complaints and generating an evidence-linked investigation output according to one or more embodiments herein. The method may be performed by the system 102 to process cybercrime complaint information, identify relationships between cybercrime complaints, determine evolutionary changes between related cybercrime patterns, and generate an investigation output supported by corresponding evidence.

[000121] At step 702, a cybercrime complaint and associated digital evidence are received. The cybercrime complaint may include information relating to a reported cybercrime incident, including complaint details, incident information, communication information, financial transaction information, and information identifying entities associated with the reported incident. The associated digital evidence may include, for example, screenshots, communication records, messages, transaction receipts, bank statements, documents, URLs, account information, or other digital information supporting the cybercrime complaint.

[000122] At step 704, the cybercrime complaint and the associated digital evidence are processed to extract and normalize entities and cybercrime events and determine evidence support. The extracted entities may include persons, organizations, phone numbers, communication accounts, URLs, domain names, UPI identifiers, bank accounts, digital wallets, or other entities associated with the cybercrime complaint. The cybercrime events may represent actions, interactions, communications, transactions, or other events associated with progression of the reported cybercrime.

[000123] The extracted entities and cybercrime events may be normalized to provide consistent representations of corresponding information obtained from the cybercrime complaint and associated digital evidence. Evidence support may further be determined for the extracted entities and cybercrime events based on corresponding information available in the associated

digital evidence. Thus, the processing at step 704 establishes a structured representation of entities, cybercrime events, and supporting evidence associated with the reported cybercrime.

[000124] At step 706, behavioural intents and an attack sequence are determined from the cybercrime events. The behavioural intents may represent objectives, strategies, or intermediate intentions associated with progression of the cybercrime attack. In some embodiments, the behavioural intents may include micro-intents associated with respective stages of the attack sequence.

[000125] For example, in an authority impersonation-based cybercrime, the determined behavioural intents may include authority impersonation, establishment of legitimacy, creation of fear or urgency, credential acquisition, and financial extraction. The attack sequence may represent the temporal or logical progression of cybercrime events associated with the determined behavioural intents. Accordingly, the determined behavioural intents and attack sequence provide information regarding both the underlying behavioural strategy and the progression of the cybercrime attack.

[000126] At step 708, a Dynamic Cyber DNA fingerprint is generated based on the processed cybercrime complaint information, the associated digital evidence, the extracted and normalized entities and cybercrime events, and the determined behavioural intents and attack sequence. The Dynamic Cyber DNA fingerprint comprises interconnected genes representing at least intent characteristics, behavioural characteristics, communication characteristics, financial characteristics, temporal characteristics, infrastructure characteristics, and evidence characteristics.

[000127] The intent characteristics may represent behavioural intentions associated with the cybercrime attack. The behavioural characteristics may represent a sequence of actions used to execute the attack. The communication characteristics may represent communication channels and interaction methods associated with the attack. The financial characteristics may represent payment methods, transaction patterns, amounts, or other financial information. The temporal characteristics may represent timing, duration, and ordering of attack events. The infrastructure characteristics may represent phone numbers, URLs, domain names, UPI identifiers, accounts, or other digital infrastructure associated with the cybercrime. The evidence characteristics may represent information supporting one or more of the other characteristics.

[000128] In some embodiments, the interconnected genes preserve relationships, sequence, contextual associations, and evidence support associated with the characteristics represented in the Dynamic Cyber DNA fingerprint. The Dynamic Cyber DNA fingerprint may therefore provide a structured representation of both underlying behavioural characteristics and observable operational characteristics associated with the cybercrime complaint.

[000129] At step 710, the generated Dynamic Cyber DNA fingerprint is compared with previously generated Cyber DNA fingerprints to identify candidate relationships between cybercrime complaints. The comparison is based on direct entity relationships and behavioural similarity. The direct entity relationships may be determined based on entities shared between corresponding cybercrime complaints or other relationships between entities represented by the respective Cyber DNA fingerprints.

[000130] The behavioural similarity may be determined based on similarities between corresponding behavioural characteristics and other relevant Cyber DNA characteristics. In some embodiments, an entity relationship score and a Cyber DNA similarity score may be determined and combined to identify candidate relationships between the cybercrime complaints. The candidate relationship may therefore be identified based on a combination of direct entity-based information and similarities in the underlying behavioural or operational characteristics.

[000131] At step 712, changes are determined between corresponding characteristics of Dynamic Cyber DNA fingerprints associated with a candidate relationship, and behavioural characteristics preserved despite the changes are identified. The determined changes may include characteristics that are modified, substituted, inserted, removed, or transformed between the corresponding Dynamic Cyber DNA fingerprints.

[000132] For example, communication through SMS may change to communication through WhatsApp, an impersonated entity may change from a bank officer to an RBI officer, an OTP-related attack method may change to a remote-access technique, or a UPI-based payment mechanism may change to a digital-wallet-based mechanism. Despite such changes, underlying behavioural characteristics, such as authority-based influence, creation of fear or urgency, victim control, and financial extraction, may remain substantially preserved.

[000133] In some embodiments, a behavioural core preservation measure is determined to assess an extent to which fundamental behavioural characteristics remain preserved between the Dynamic Cyber DNA fingerprints. A context mutation measure may also be determined to assess

an extent to which contextual or operational characteristics have changed. The identified changes and preserved behavioural characteristics may further be used to determine a mutation path representing a possible progression from one Cyber DNA fingerprint to another.

[000134] At step 714, a candidate Cyber DNA evolutionary lineage is generated based on the determined changes, the preserved behavioural characteristics, temporal information, and supporting evidence. The candidate Cyber DNA evolutionary lineage may organize related Cyber DNA fingerprints according to their temporal occurrence and candidate evolutionary relationships.

[000135] In some embodiments, the candidate evolutionary relationship may be evaluated based on structural similarity between corresponding Cyber DNA fingerprints, mutation cost associated with the determined changes, behavioural core preservation, temporal plausibility, and evidence support. The resulting candidate Cyber DNA evolutionary lineage may represent a possible evolution of a cybercrime pattern in which operational characteristics change over time while significant portions of an underlying behavioural strategy remain preserved.

[000136] The candidate Cyber DNA evolutionary lineage represents an analytical relationship between related cybercrime patterns and does not, by itself, establish that the same offender or criminal organization performed the respective cybercrime attacks.

[000137] At step 716, an evidence-linked investigation output is generated identifying the candidate relationship, the determined changes, the preserved behavioural characteristics, and the candidate Cyber DNA evolutionary lineage. The evidence-linked investigation output may further identify corresponding entities, cybercrime events, mutation information, temporal information, and supporting evidence associated with the generated analytical results.

[000138] In some embodiments, the evidence-linked investigation output enables an investigator to review the basis for the identified candidate relationship and the candidate Cyber DNA evolutionary lineage. The output may preserve associations between the identified findings and corresponding supporting evidence, thereby providing an explainable representation of the relationship between the cybercrime complaints and the determined evolutionary changes.

[000139] The embodiments herein provide several technical advantages. The system enables generation of a Dynamic Cyber DNA fingerprint that preserves interconnected intent, behavioural, communication, financial, temporal, infrastructure, and evidence characteristics associated with a cybercrime complaint, thereby providing a structured representation of the

cybercrime pattern. The system further enables identification of candidate relationships by considering both direct entity relationships and behavioural similarity, reducing reliance on isolated matching of individual entities. By determining changes between corresponding characteristics while identifying behavioural characteristics preserved despite such changes, the system enables identification and analysis of evolving cybercrime patterns. Further, the generation of a candidate Cyber DNA evolutionary lineage based on changes, preserved behavioural characteristics, temporal information, and supporting evidence enables structured analysis of relationships among related cybercrime complaints. The evidence-linked investigation output further improves traceability by associating identified relationships, changes, preserved behavioural characteristics, and evolutionary information with supporting evidence.

[000140] The foregoing description of the specific embodiments will so fully reveal the general nature of the embodiments herein that others can, by applying current knowledge, readily modify and/or adapt for various applications such specific embodiments without departing from the generic concept, and, therefore, such adaptations and modifications should and are intended to be comprehended within the meaning and range of equivalents of the disclosed embodiments. It is to be understood that the phraseology or terminology employed herein is for the purpose of description and not of limitation. Therefore, while the embodiments herein have been described in terms of preferred embodiments, those skilled in the art will recognize that the embodiments herein can be practiced with modification within the spirit and scope of the appended claims.

## CLAIMS

I/ We claim:

1. A system (102) for dynamic Cyber DNA fingerprinting and evolutionary analysis of cybercrime complaints, the system (102) comprising:

a memory (102B) configured to store cybercrime complaints, associated digital evidence, previously generated Cyber DNA fingerprints, and historical cybercrime relationship information; and

one or more processors (102A) communicatively coupled to the memory (102B) and configured to:

receive a cybercrime complaint and associated digital evidence comprising one or more of text, images, screenshots, messages, emails, chat records, transaction records, bank statements, URLs, QR codes, documents, and audio recordings;

process the cybercrime complaint and the associated digital evidence to extract and normalize entities and cybercrime events, and determine evidence support for the extracted and normalized entities and cybercrime events;

determine behavioural intents and an attack sequence from the extracted cybercrime events;

generate, based on the extracted and normalized entities, cybercrime events, behavioural intents, attack sequence, evidence support, and temporal information, a Dynamic Cyber DNA fingerprint comprising interconnected genes representing at least intent characteristics, behavioural characteristics, communication characteristics, financial characteristics, temporal characteristics, infrastructure characteristics, and evidence characteristics;

compare the Dynamic Cyber DNA fingerprint with previously generated Cyber DNA fingerprints based on direct entity relationships and behavioural similarity to identify candidate relationships between cybercrime complaints;

determine, for a candidate relationship, changes between corresponding characteristics of the Dynamic Cyber DNA fingerprints and behavioural characteristics preserved despite the changes;

generate a candidate Cyber DNA evolutionary lineage based on the determined changes, the preserved behavioural characteristics, temporal information, and evidence supporting the candidate relationship; and

generate an evidence-linked investigation output identifying the candidate relationship, the determined changes, the preserved behavioural characteristics, and the candidate Cyber DNA evolutionary lineage.

2. The system (102) as claimed in claim 1, wherein the one or more processors (102A) are configured to process each item of the associated digital evidence according to an evidence type of the item and to generate a structured evidence intelligence record comprising normalized entities, cybercrime events, source provenance, extraction confidence, evidence agreement information, and conflicting evidence information.

3. The system (102) as claimed in claim 1, wherein the one or more processors (102A) are configured to determine the behavioural intents by identifying respective micro-intents associated with the cybercrime events and arranging the respective micro-intents according to the attack sequence, each micro-intent being associated with corresponding supporting evidence and a confidence value based on semantic correspondence, evidence support, and contextual consistency.

4. The system (102) as claimed in claim 1, wherein each gene of the Dynamic Cyber DNA fingerprint comprises a gene type, an encoded characteristic, an associated confidence value, a temporal position, and one or more relationships with other genes, thereby preserving a sequence and interrelationship of characteristics associated with the cybercrime complaint.

5. The system (102) as claimed in claim 1, wherein the one or more processors (102A) are configured to update the Dynamic Cyber DNA fingerprint upon receipt of additional evidence or validated investigative information by identifying one or more genes affected by the additional evidence or validated investigative information, updating the identified genes while retaining unaffected genes, and maintaining a previous version and an updated version of the Dynamic Cyber DNA fingerprint with corresponding evidence provenance and update information.

6. The system (102) as claimed in claim 1, wherein identifying the candidate relationship comprises determining an entity relationship score based on shared entities and an adaptive reliability weight associated with each entity type, determining a Cyber DNA similarity score based on similarities between corresponding genes of the Dynamic Cyber DNA fingerprints, and combining the entity relationship score and the Cyber DNA similarity score to generate a hybrid relationship score.

7. The system (102) as claimed in claim 1, wherein determining the changes and the preserved behavioural characteristics comprises identifying one or more characteristics that are preserved, modified, substituted, inserted, removed, or transformed between the Dynamic Cyber DNA fingerprints, and determining a behavioural core preservation measure and a context mutation measure to distinguish a candidate evolved variant from a merely similar cybercrime complaint.

8. The system (102) as claimed in claim 1, wherein the one or more processors (102A) are further configured to store sufficiently supported or investigator-validated transformations between Cyber DNA characteristics in a Crime Evolution Memory, group related Dynamic Cyber DNA fingerprints into a candidate campaign profile, and estimate one or more candidate future variants based on previously stored transformations and contextual transition information, each candidate future variant being associated with supporting historical transition information and a probability or confidence value.

9. A computer-implemented method for dynamic Cyber DNA fingerprinting and evolutionary analysis of cybercrime complaints, the method comprising:

receiving a cybercrime complaint and associated digital evidence;

processing the cybercrime complaint and the associated digital evidence to extract and normalize entities and cybercrime events and determine evidence support;

determining behavioural intents and an attack sequence from the cybercrime events;

generating a Dynamic Cyber DNA fingerprint comprising interconnected genes representing at least intent characteristics, behavioural characteristics, communication

characteristics, financial characteristics, temporal characteristics, infrastructure characteristics, and evidence characteristics;

comparing the Dynamic Cyber DNA fingerprint with previously generated Cyber DNA fingerprints based on direct entity relationships and behavioural similarity to identify candidate relationships between cybercrime complaints;

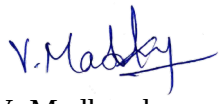
determining changes between corresponding characteristics of Dynamic Cyber DNA fingerprints associated with a candidate relationship and behavioural characteristics preserved despite the changes;

generating a candidate Cyber DNA evolutionary lineage based on the changes, the preserved behavioural characteristics, temporal information, and supporting evidence; and

generating an evidence-linked investigation output identifying the candidate relationship, the changes, the preserved behavioural characteristics, and the candidate Cyber DNA evolutionary lineage.

10. The method as claimed in claim 9, further comprising receiving investigator feedback indicating that a candidate relationship, mutation relationship, or campaign association is confirmed, rejected, or requires further investigation, and responsively updating one or more of entity reliability weights, Cyber DNA similarity parameters, mutation relationships, evolutionary lineage assessment parameters, and stored historical transformation information while preserving original evidence, generated Cyber DNA versions, and historical analytical results.

Dated this 28th August, 2026

Signature: 

Name: V. Madhan kumar

(IN/PA –4321)

ABSTRACT

SYSTEM AND METHOD FOR DYNAMIC CYBER DNA FINGERPRINTING AND EVOLUTIONARY ANALYSIS OF CYBERCRIME COMPLAINTS

A system (102) for dynamic Cyber DNA fingerprinting and evolutionary analysis of cybercrime complaints is disclosed. The system (102) receives a cybercrime complaint and associated digital evidence, processes the complaint and evidence to extract and normalize entities and cybercrime events, and determines evidence support. Behavioural intents and an attack sequence are determined from the cybercrime events. A Dynamic Cyber DNA fingerprint comprising interconnected genes representing intent, behavioural, communication, financial, temporal, infrastructure, and evidence characteristics is generated. The fingerprint is compared with previously generated Cyber DNA fingerprints based on direct entity relationships and behavioural similarity to identify candidate relationships. Changes between corresponding characteristics and preserved behavioural characteristics are determined. A candidate Cyber DNA evolutionary lineage is generated based on the changes, preserved behavioural characteristics, temporal information, and supporting evidence. An evidence-linked investigation output is generated identifying the candidate relationship, changes, preserved behavioural characteristics, and evolutionary lineage.

FIG. 1

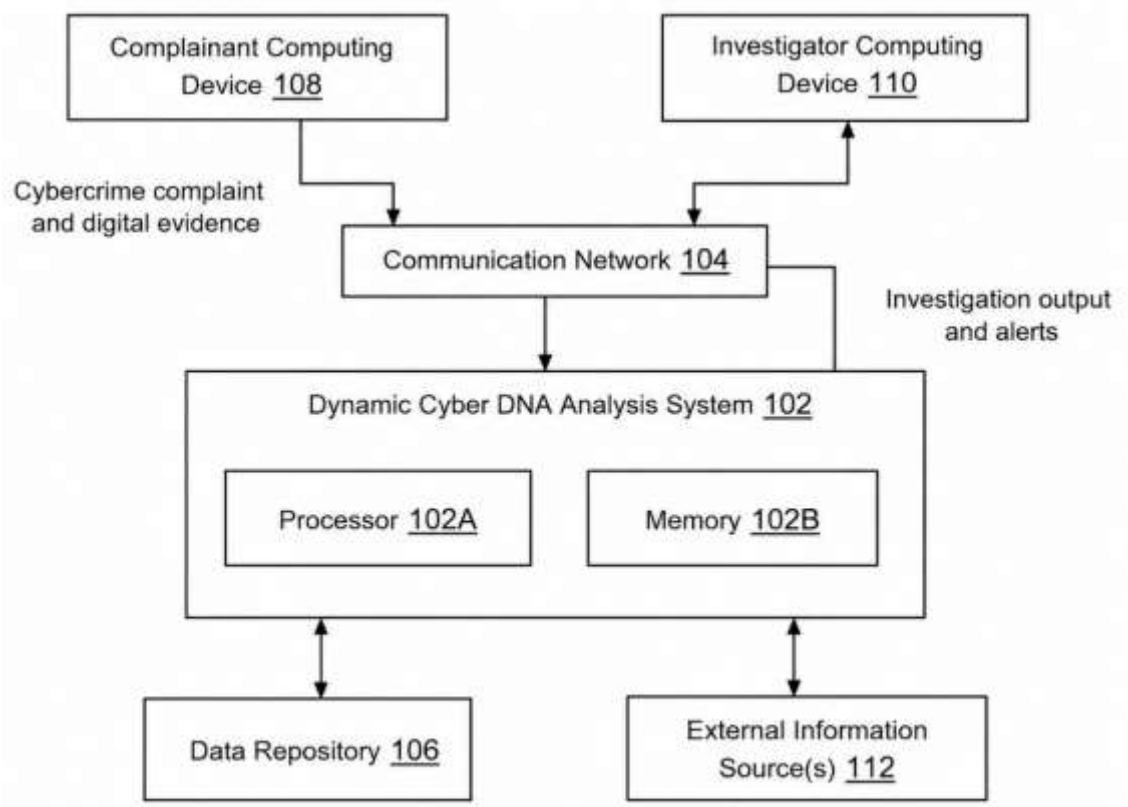



FIG. 1

SIGNATURE OF THE AGENT:

NAME: V. MADHAN KUMAR
IN/PA –4321

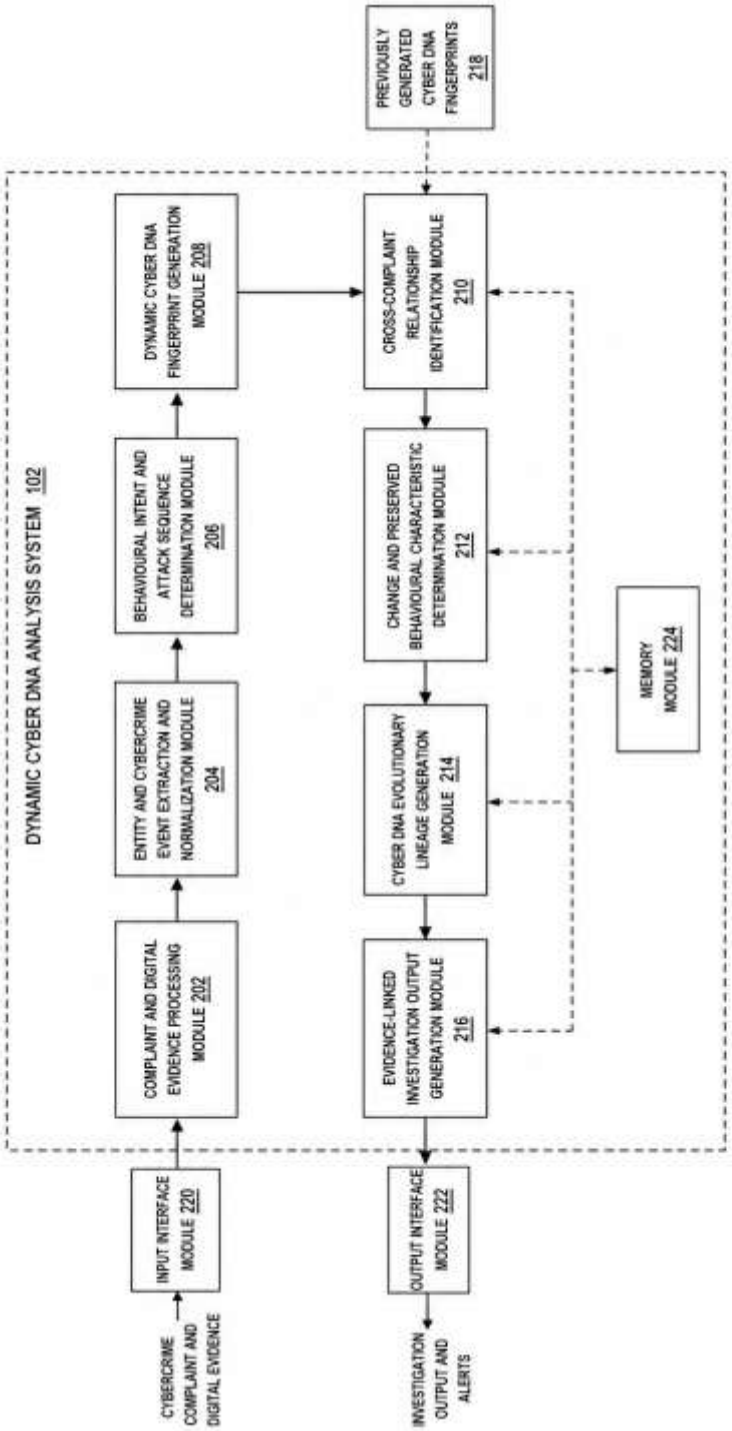



FIG. 2

SIGNATURE OF THE AGENT:

NAME: V. MADHAN KUMAR

IN/PA –4321

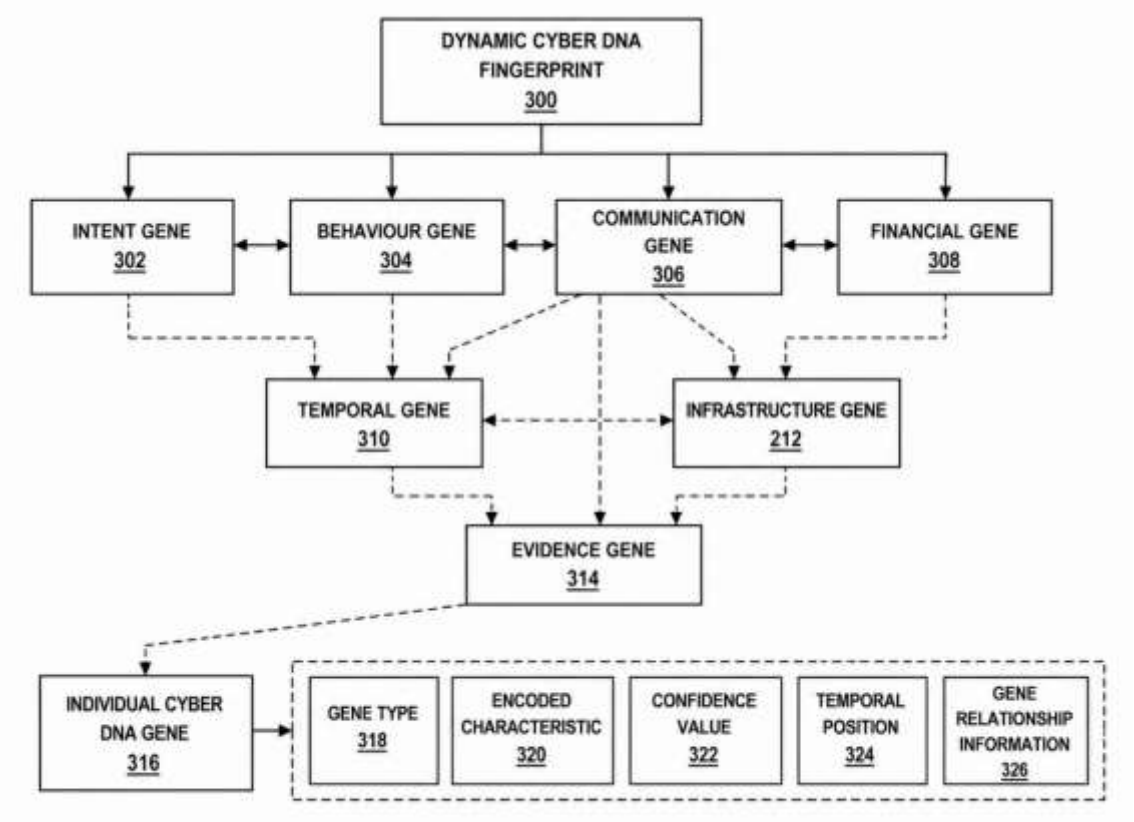



FIG. 3

SIGNATURE OF THE AGENT:

NAME: V. MADHAN KUMAR
IN/PA –4321

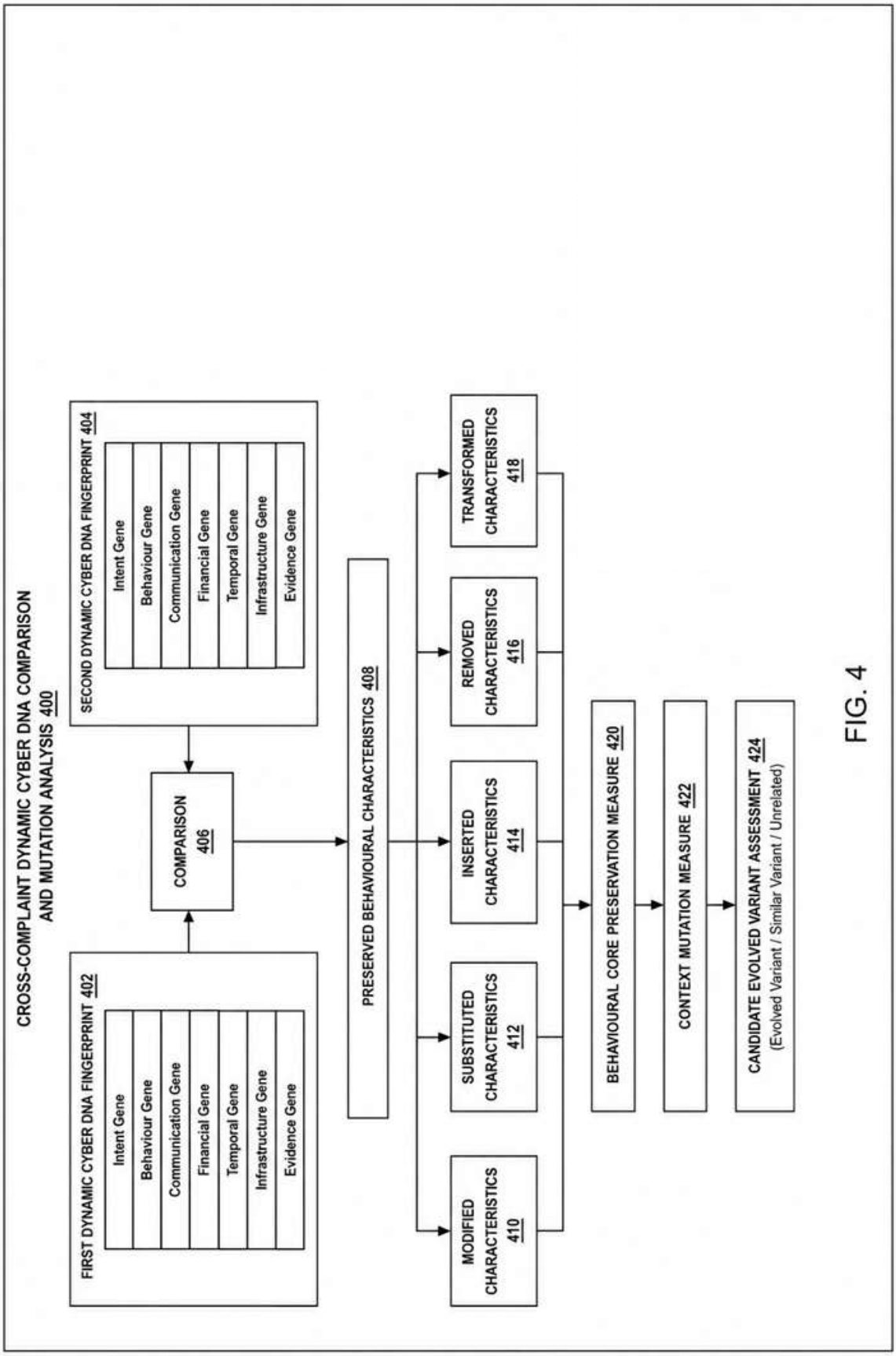



FIG. 4

SIGNATURE OF THE AGENT:

NAME: V. MADHAN KUMAR
IN/PA –4321

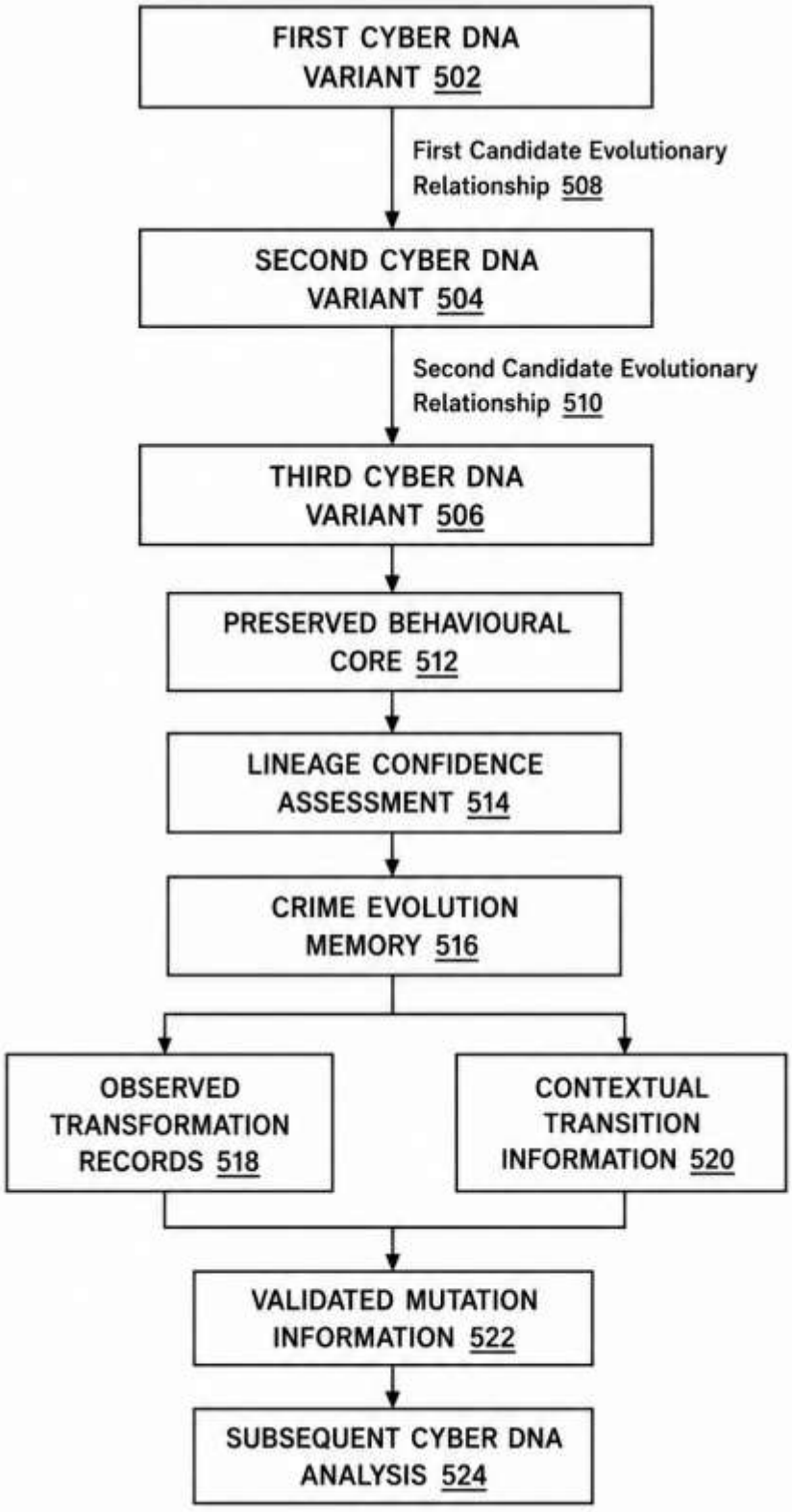



FIG. 5

SIGNATURE OF THE AGENT:

NAME: V. MADHAN KUMAR
IN/PA –4321

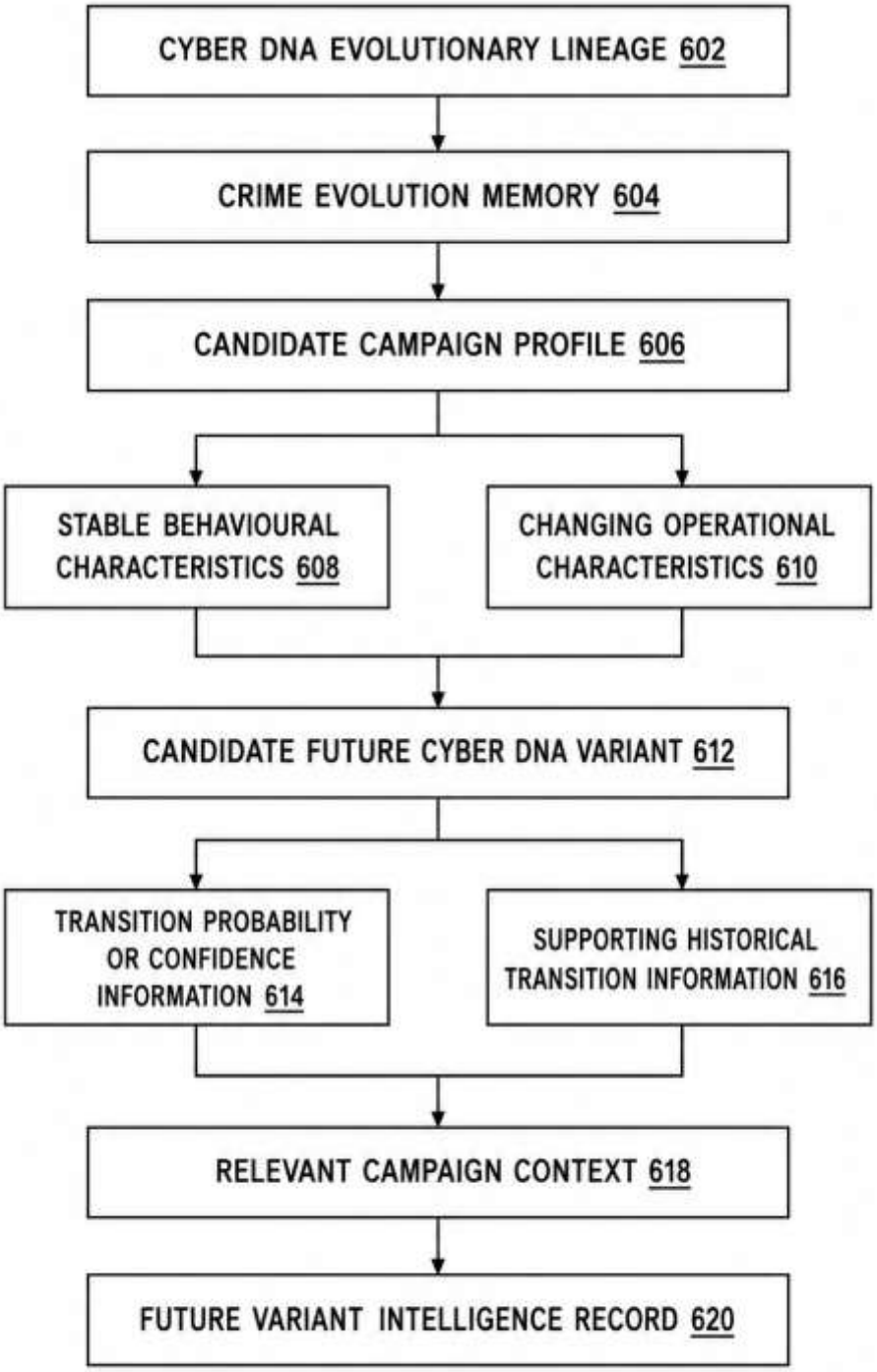



FIG. 6

SIGNATURE OF THE AGENT:

NAME: V. MADHAN KUMAR
IN/PA –4321

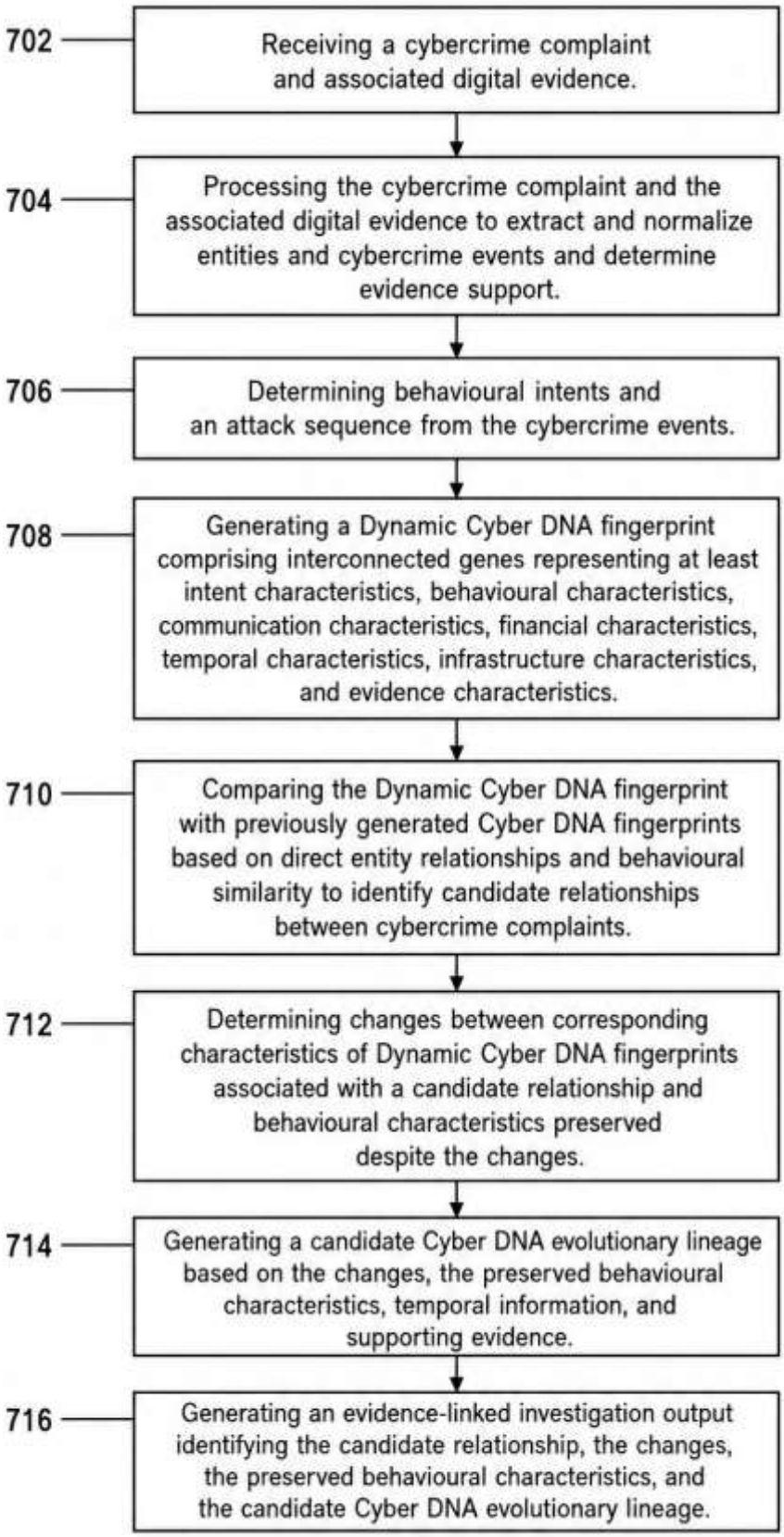



FIG. 7

SIGNATURE OF THE AGENT:

NAME: V. MADHAN KUMAR
IN/PA –4321

Welcome MADHAN KUMAR Sign out

**Controller General of Patents, Designs & Trade Marks**

सत्यमेव जयते

INTELLECTUAL PROPERTY INDIA
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GEOGRAPHICAL INDICATIONS

**G.A.R.6**
**[See Rule 22(1)]**
**RECEIPT**

**Docket No 142308** **Date/Time 2026/09/10 07:19:23**

**MADHAN KUMAR 29,**
**Nandhanam,Vishal Garden,**
**Suleeshwaranpatti, Pollachi Email:**
**kmipconsultancy@gmail.com**

| Sr. No. | App. Number | Ref. No./Application No. | Amount Paid | C.B.R. No. | Form Name | Fee Payment | Remarks |
|---|---|---|---|---|---|---|---|
| 1 | 202641108533 | TEMP/E-1/114754/2026-CHE | 1600 | 73803 | FORM 1 | Full | CRYPTOGRAPHIC HARDWARE SYSTEM AND METHOD FOR SECURE TELEMETRY COMMUNICATION IN IMPLANTABLE MEDICAL DEVICE |
| 2 | E-106/23289/2026/CHE | 202641108528 | 0 | -1 | FORM28 | Full | |
| 3 | 202641108530 | TEMP/E-1/116350/2026-CHE | 4160 | 73803 | FORM 1 | Full | SYSTEM AND METHOD FOR DYNAMIC CYBER DNA FINGERPRINTING AND EVOLUTIONARY ANALYSIS OF CYBERCRIME COMPLAINTS |
| 4 | 202641108528 | TEMP/E-1/114731/2026-CHE | 2560 | 73803 | FORM 1 | Full | SYSTEM AND METHOD FOR CORROBORATING EMERGENCY EVENTS IN AN INFRASTRUCTURE-INDEPENDENT WIRELESS NETWORK |
| 5 | E-106/23287/2026/CHE | 202641108529 | 0 | -1 | FORM28 | Full | |
| 6 | E-106/23295/2026/CHE | 202641108530 | 0 | -1 | FORM28 | Full | |
| 7 | 202641108525 | TEMP/E-1/114787/2026-CHE | 1600 | 73803 | FORM 1 | Full | SYSTEM AND METHOD FOR HARDWARE-AWARE CONTEXT TRANSFORMATION |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | | | | | AND MULTI-TIER MEMORY ORCHESTRATION FOR LONG-CONTEXT INFERENCE |
| 8 | E-106/23293/2026/CHE | 202641108533 | 0 | -1 | FORM28 | Full | |
| 9 | E-12/29490/2026/CHE | 202641108529 | 2500 | 73803 | FORM 9 | Full | |
| 10 | E-106/23292/2026/CHE | 202641108527 | 0 | -1 | FORM28 | Full | |
| 11 | 202641108526 | TEMP/E-1/114826/2026-CHE | 3360 | 73803 | FORM 1 | Full | SYSTEM AND METHOD FOR DYNAMIC TRUST-BASED SELECTION OF SERVERLESS FUNCTION INSTANCES |
| 12 | 202641108527 | TEMP/E-1/117044/2026-CHE | 1760 | 73803 | FORM 1 | Full | SYSTEM AND METHOD FOR ENTROPY-DRIVEN DYNAMIC WORKLOAD ORCHESTRATION IN HYBRID QUANTUM-CLASSICAL METEOROLOGICAL FORECASTING |
| 13 | E-106/23290/2026/CHE | 202641108525 | 0 | -1 | FORM28 | Full | |
| 14 | E-12/29487/2026/CHE | 202641108528 | 2500 | 73803 | FORM 9 | Full | |
| 15 | E-12/29495/2026/CHE | 202641108527 | 2500 | 73803 | FORM 9 | Full | |
| 16 | E-12/29491/2026/CHE | 202641108526 | 2500 | 73803 | FORM 9 | Full | |
| 17 | E-12/29488/2026/CHE | 202641108531 | 2500 | 73803 | FORM 9 | Full | |
| 18 | E-106/23288/2026/CHE | 202641108532 | 0 | -1 | FORM28 | Full | |
| 19 | E-12/29494/2026/CHE | 202641108525 | 2500 | 73803 | FORM 9 | Full | |
| 20 | 202641108532 | TEMP/E-1/119460/2026-CHE | 2880 | 73803 | FORM 1 | Full | CONDUCTOR STRUCTURAL SIGNATURE-BASED ELECTRICITY THEFT IDENTIFICATION SYSTEM AND METHOD |
| 21 | 202641108529 | TEMP/E-1/114699/2026-CHE | 2720 | 73803 | FORM 1 | Full | SYSTEM AND METHOD FOR LOW-LATENCY SEMANTIC COMMUNICATION BETWEEN HETEROGENEOUS NEURAL NETWORKS |
| 22 | E-12/29493/2026/CHE | 202641108532 | 2500 | 73803 | FORM 9 | Full | |
| 23 | E-12/29492/2026/CHE | 202641108530 | 2500 | 73803 | FORM 9 | Full | |
| 24 | E-106/23294/2026/CHE | 202641108531 | 0 | -1 | FORM28 | Full | |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 25 | E-106/23291/2026/CHE | 202641108526 | 0 | -1 | FORM28 | Full | |
| 26 | E-12/29489/2026/CHE | 202641108533 | 2500 | 73803 | FORM 9 | Full | |
| 27 | 202641108531 | TEMP/E-1/119441/2026-CHE | 3680 | 73803 | FORM 1 | Full | SYSTEM AND METHOD FOR CLOSED-LOOP INFERENCE CONTROL AND DYNAMIC THRESHOLD RECALIBRATION OF TRANSFORMER-BASED LANGUAGE MODELS |

| TransactionID | Payment Mode | Challan Identification Number | Amount Paid | Head of A/C No |
|---|---|---|---|---|
| N-0002052936 | Online Bank Transfer | 1009260002509 | 46820.00 | 1475001020000001 |

Total Amount : ₹ 46820.00

Amount in Words: Rupees Forty Six Thousand Eight Hundred Twenty Only

Received from MADHAN KUMAR the sum of ₹ 46820.00 on account of Payment of fee for above mentioned Application/Forms.

* This is a computer generated receipt, hence no signature required.

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